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AUO-General

Product Specification

AU OPTRONICS CORPORATION

( ) Preliminary Specifications
(V) Final Specifications

Module	**16.0”QHD+ 16:10 Color TFT-LCD with LED Backlight design**
Model Name	B160QAN02.Q
H/W	**0A**
Note ( )	***LED Backlight with driving circuit design***

Customer	Date	Approved by	Date
ASUS	05/14/2021	Randolph Chang	05/14/2021
Checked & Approved by	**Date**	**Prepared by**	**Date**
XXXXXX	MM/DD/YYYY	Claire Yu	05/14/2021
Note: This Specification is subject to change without notice.		**AU Optronics corporation**	

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Record of Revision

Version and Date	Page	Old description	New Description	Remark
0.1 2021/12/29	All	First Edition for Customer		
1.0 2021/02/19	6		Modified 13P Uniformity & color	
	22		Modified 165Hz Timing	
	28		Modified Back side drawing	
	29		Label information	
	31-36		Modified EDID	
	37		Notes (add timing3)	
1.1 2021/03/15	29-30		Update Carton packing 18pcs→24pcs	
1.2 2021/04/13	All	Final Edition for Customer		
1.3 2021/05/05	31-34		Update EDID(for R/G/B color spec)	
1.4 2021/05/14	13		Update 5.1.1, heavy pattern power	

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1. Handling Precautions

1) Since front polarizer is easily damaged, pay attention not to scratch it.
2) Be sure to turn off power supply when inserting or disconnecting from input connector.
3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
4) When the panel surface is soiled, wipe it with absorbent cotton or other soft cloth.
5) Since the panel is made of glass, it may break or crack if dropped or bumped on hard surface.
6) Since CMOS LSI is used in this module, take care of static electricity and insure human earth when handling.
7) Do not open nor modify the Module Assembly.
8) Do not press the reflector sheet at the back of the module to any directions.
9) At the insertion or removal of the Signal Interface Connector, be sure not to rotate nor tilt the Interface Connector of the TFT Module.
11) After installation of the TFT Module into an enclosure (Notebook PC Bezel, for example), do not twist nor bend the TFT Module even momentary. At designing the enclosure, it should be taken into consideration that no bending/twisting forces are applied to the TFT Module from outside. Otherwise the TFT Module may be damaged.
12) Small amount of materials having no flammability grade is used in the LCD module. The LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950 or UL1950), or be applied exemption.
13) Disconnecting power supply before handling LCD modules, it can prevent electric shock, DO NOT TOUCH the electrode parts, cables, connectors and LED circuit part of TFT module that a LED light bar build in as a light source of back light unit. It can prevent electrostatic breakdown.

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2. General Description

B160QAN02.Q is a Color Active Matrix Liquid Crystal Display composed of a TFT LCD panel, a driver circuit, and LED backlight system. The screen format is intended to support the 16:10 QHD+, 2560(H) x1600(V) screen and 16.7M colors (RGB 8-bits data driver) with LED backlight driving circuit. All input signals are eDP(Embedded DisplayPort) interface compatible.

B160QAN02.Q is designed for a display unit of notebook style personal computer and industrial machine.

2.1 General Specification

The following items are characteristics summary on the table at 25 ℃ condition:

Items	Unit	Specifications			
Screen Diagonal	[in]	16.0”(16”)			
Active Area	[mm]	344.68 X 215.42			
Pixels H x V		2560x3(RGB) x 1600			
Pixel Pitch	[mm]	0.1346 x 0.1346			
Pixel Format		R.G.B. Vertical Stripe			
Display Mode		Normally Black (AHVA)			
White Luminance (I_{LED}=20mA) **(Note: ILED is LED current)**	[cd/m^2]	500 typ. (5 points average) 425 min. (5 points average)			
Luminance Uniformity		1.25 max. (5 points)			
Contrast Ratio		1000 typ. 800min			
Response Time	[ms]	G2G 5ms			
Nominal Input Voltage VDD	[Volt]	+3.3 typ.			
Power Consumption	[Watt]	7.6 max. (Include Logic and Blu power)			
Weight	[Grams]	330 max.			
Physical Size **Include PCBA** (refer 2D drawing)	[mm]		Min.	Typ.	Max.
		Length	349.378	349.678	349.978
		Width		244.52	224.924
		Thickness	-	-	2.6/4.6
Electrical Interface		4 Lane eDP 1.4			
Glass Thickness	[mm]	0.25			
Surface Treatment		Anti-glare,			
Support Color		16.7M colors (RGB 8-bit)			

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Temperature Range Operating Storage (Non-Operating)	 [°C] [°C]	 0 to +50 -20 to +60
RoHS Compliance		RoHS Compliance

2.2 Optical Characteristics

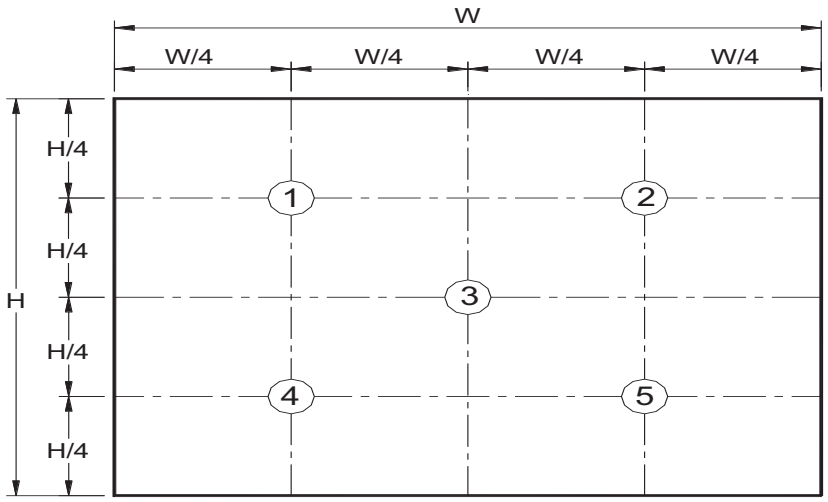
The optical characteristics are measured under stable conditions at 25℃ (Room Temperature) :

Item		Symbol	Conditions	Min.	Typ.	Max.	Unit	Note
White Luminance I_{LED}=20mA		Y_L	5 points average	425	500	-	cd/m²	1, 4, 5.
Viewing Angle		θ_R θ_L	Horizontal (Right) CR = 10 (Left)	80 80	85 85	- -	degree	4, 9
		ϕ_H ϕ_L	Vertical (Upper) CR = 10 (Lower)	80 80	85 85	- -		
Luminance Uniformity		δ_{5P}	5 Points	-	-	1.25		1, 3, 4
Luminance Uniformity		δ_{13P}	13 Points	-	-	1.6		2, 3, 4
Contrast Ratio		CR		800	1000	-		4, 6
Cross talk		%				4		4, 7
Response Time		T_{RT}	Rising + Falling	-	8	16	msec	4, 8
Response Time		$T_{G\ To\ G}$			5		msec	8
Response Time		$T_{OD\ G\ To\ G}$			3		msec	8
Color / Chromaticity Coordinates	Red	Rx	CIE 1931	0.649	0.679	0.709		Might be updated before MP
		Ry		0.286	0.316	0.346		
	Green	Gx		0.228	0.258	0.288		
		Gy		0.663	0.693	0.723		
	Blue	Bx		0.117	0.147	0.177		
		By		0.022	0.052	0.082		
	White	Wx		0.283	0.313	0.343		
		Wy		0.299	0.329	0.359		
DCI-P3		%		95	100			4
Gamma Value		-		2.0	2.2	2.4		Follow AUO's spec

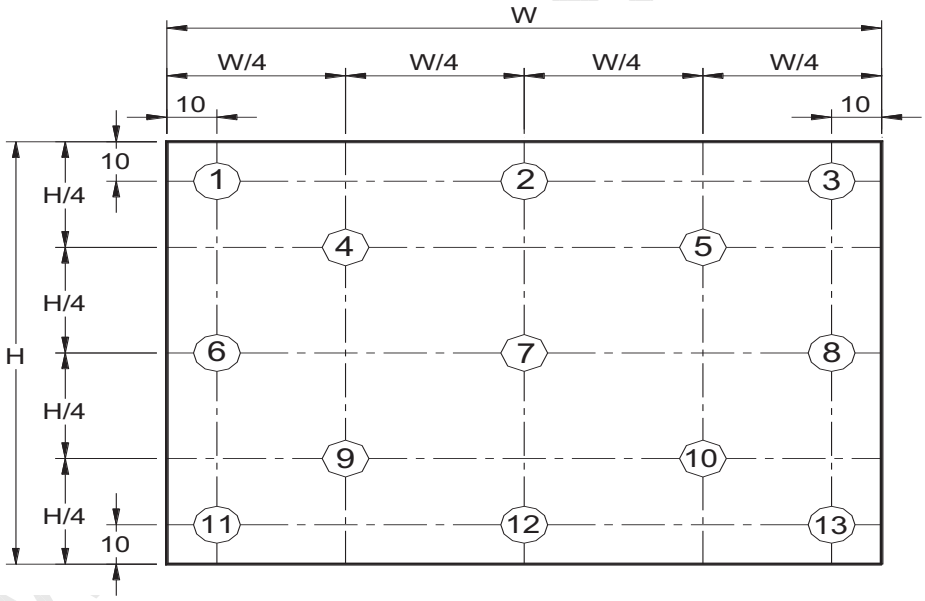
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Note 1: 5 points position (Ref: Active area)



Note 2: 13 points position (Ref: Active area)



Note 3: The luminance uniformity of 5 or13 points is defined by dividing the maximum luminance values by the minimum test point luminance

$$\delta_{W5} = \frac{\text{Maximum Brightness of five points}}{\text{Minimum Brightness of five points}}$$

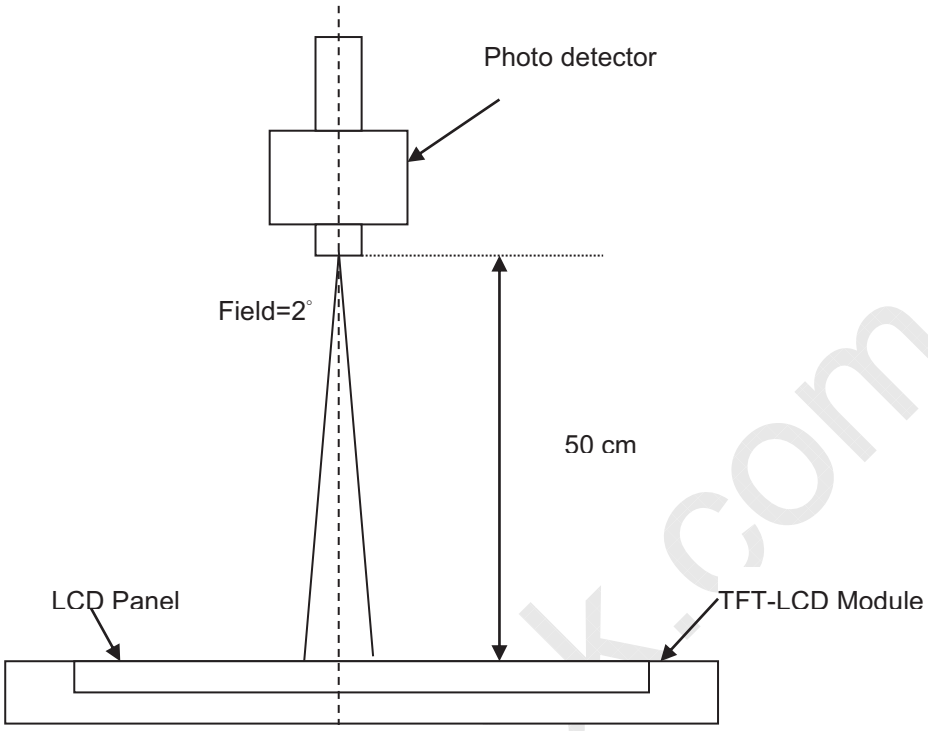
$$\delta_{W13} = \frac{\text{Maximum Brightness of thirteen points}}{\text{Minimum Brightness of thirteen points}}$$

Note 4: Measurement method

The LCD module should be stabilized at given temperature for 30 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 30 minutes in a stable, windless and dark room, and it should be measured in the center of screen.

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Note 5： Definition of Average Luminance of White (Y_L):

Measure the luminance of gray level 63 at 5 points，Y_L = [L (1)+ L (2)+ L (3)+ L (4)+ L (5)] / 5

L (x) is corresponding to the luminance of the point X at Figure in Note (1).

Note 6： Definition of contrast ratio:

Contrast ratio is calculated with the following formula.

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness on the “White” state}}{\text{Brightness on the “Black” state}}$$

Note 7： Definition of Cross Talk (CT)

$CT = |Y_B - Y_A| / Y_A \times 100\ (\%)$

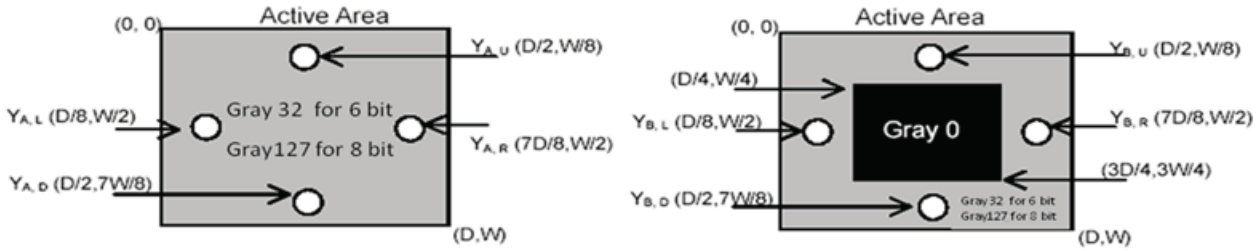
Where

Y_A = Luminance of measured location without gray level 0 pattern (cd/m^2)

Y_B = Luminance of measured location with gray level 0 pattern (cd/m^2)

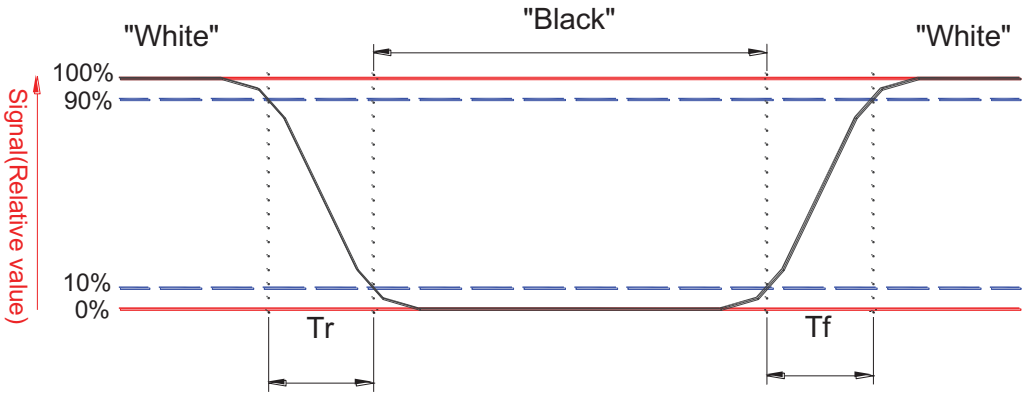
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Note 8: Definition of response time:

The output signals of BM-7 or equivalent are measured when the input signals are changed from “Black” to “White” (falling time) and from “White” to “Black” (rising time), respectively. The response time interval between the 10% and 90% of amplitudes. Refer to figure as below.



The gray to gray response time is defined as the following table.

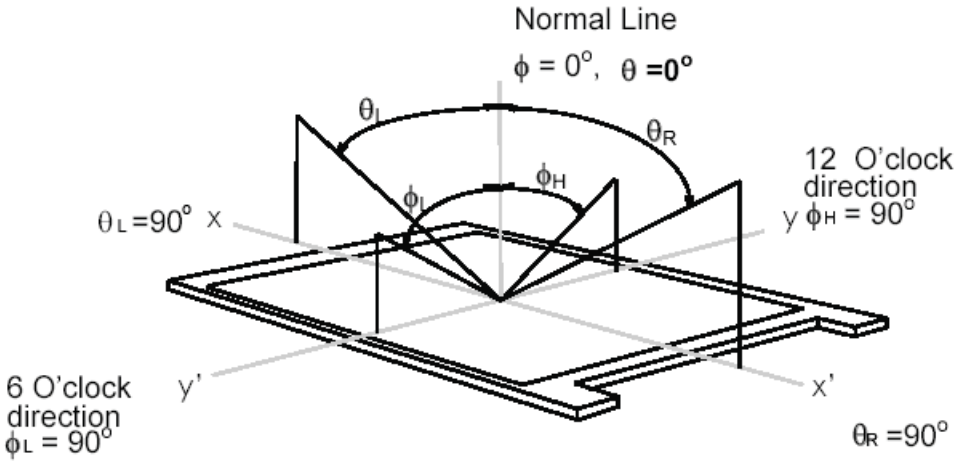
Response Time9*9 matrix										
Response Time		To								
		L0	L32	L64	L96	L128	L160	L192	L224	L255
From	L0									
	L32									
	L64									
	L96									
	L128									
	L160									
	L192									
	L224									
	L255									
Response time (Tr+Tf)=L0 to L255+L255 to L0										
Response time (Gray to Gray) average = average time in 9*9 matrix										

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Note 9. Definition of viewing angle

Viewing angle is the measurement of contrast ratio ≧10, at the screen center, over a 180° horizontal and 180° vertical range (off-normal viewing angles). The 180° viewing angle range is broken down as follows; 90° (θ) horizontal left and right and 90° (Φ) vertical, high (up) and low (down). The measurement direction is typically perpendicular to the display surface with the screen rotated about its center to develop the desired measurement viewing angle.

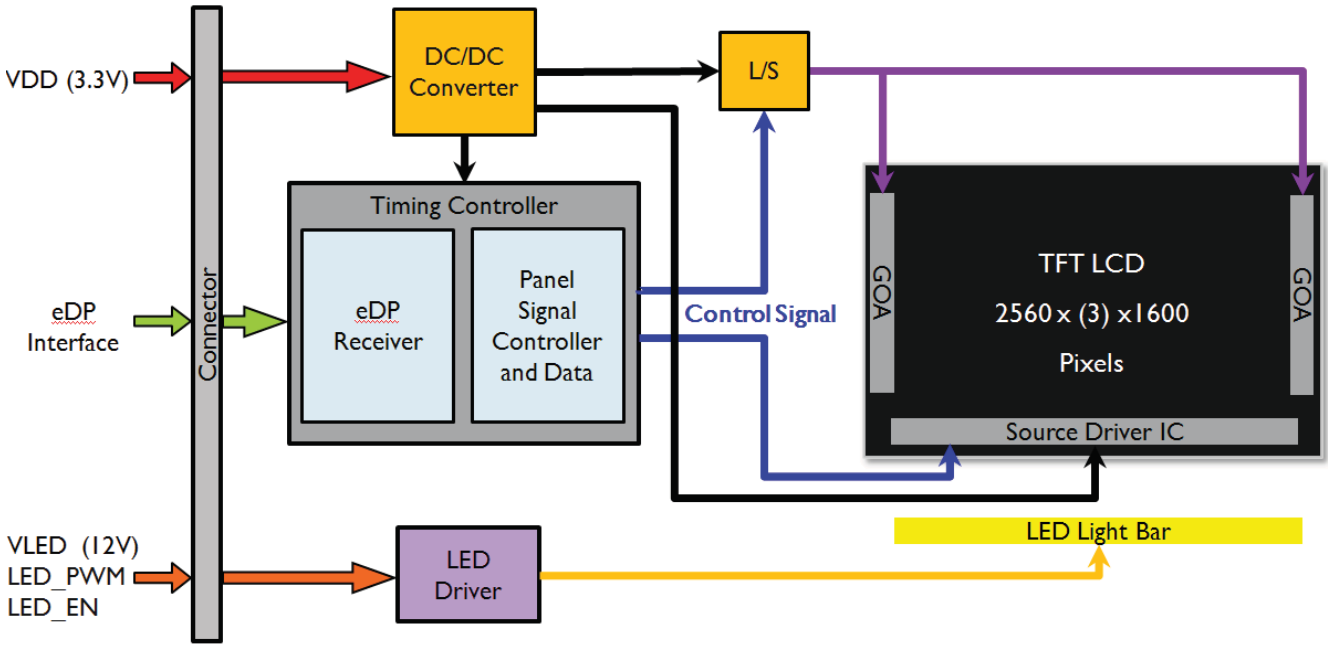


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3. Functional Block Diagram

The following diagram shows the functional block of the 16.0 inches wide Color TFT/LCD 40 Pin



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4. Absolute Maximum Ratings

An absolute maximum rating of the module is as following:

4.1 Absolute Ratings of TFT LCD Module

Item	Symbol	Min	Max	Unit	Conditions
Logic/LCD Drive Voltage	Vin	-0.3	+4.0	[Volt]	Note 1,2

4.2 Absolute Ratings of Environment

Item	Symbol	Min	Max	Unit	Conditions
Operating Temperature	TOP	0	+50	[°C]	Note 4
Operating Humidity	HOP	5	90	[%RH]	Note 4
Storage Temperature	TST	-20	+60	[°C]	Note 4
Storage Humidity	HST	5	90	[%RH]	Note 4

Note 1: At Ta (25℃)

Note 2: Permanent damage to the device may occur if exceed maximum values

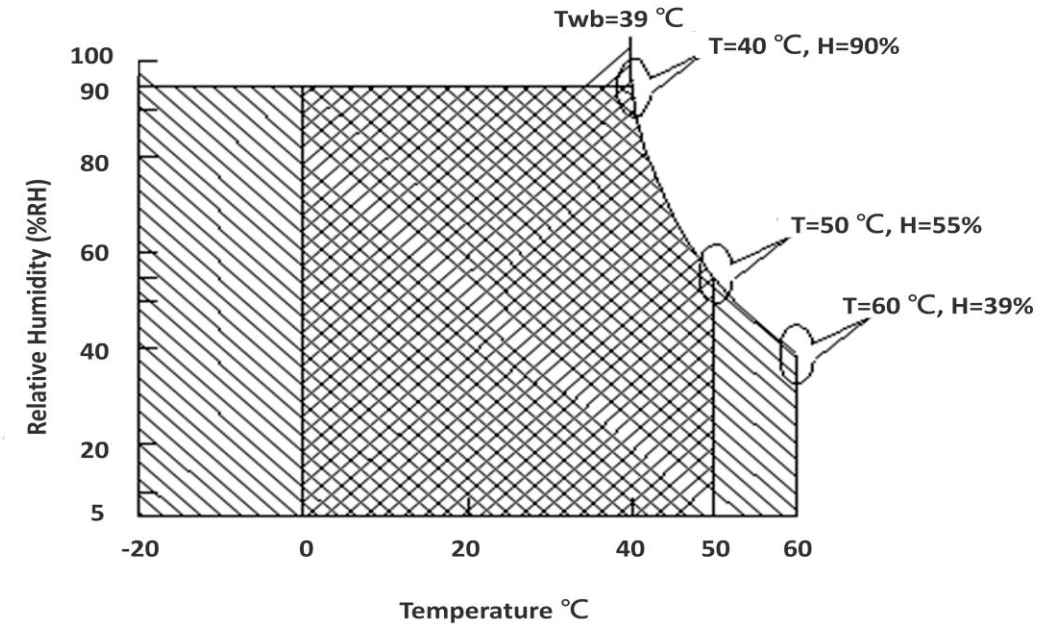
Note 3: LED specification refer to section 5.2

Note 4: For quality performance, please refer to AUO IIS (Incoming Inspection Standard).

Note 5: The packing material of system forbid to involve ammonium component

Note 6: The reliability test conditions of system do not exceed the verified conditions of TFT module

Note 7: Be sure the panel test condition do not exceed the component limitation of TFT module(TN Liquid crystal , for example)



Operating Range Storage Range

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5. Electrical Characteristics

5.1 TFT LCD Module

5.1.1 Power Specification

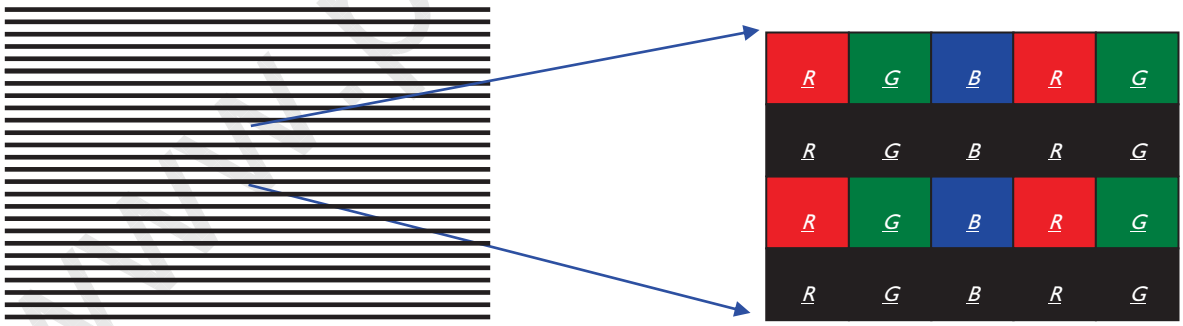
Input power specifications are as follows;

The power specification are measured under 25℃ and frame frequency under 165Hz

Symble	Parameter		Min	Typ	Max	Units	Note
VDD	Logic/LCD Drive Voltage		3.0	3.3	3.6	[Volt]	
PDD	VDD Power	Mosaic		1.5	2.0	[Watt]	Note 1
		White		1.5	2.0	[Watt]	
		Red		1.5	2.0	[Watt]	
		H-stripe		3.0	4.0	[Watt]	Note 2
IDD	IDD Current	Mosaic		455	667	[mA]	
		White		455	667	[mA]	
		Red		455	667	[mA]	
		H-stripe		910	1333	[mA]	
IRush	Inrush Current				2000	[mA]	Note 3
VDDrp	Allowable Logic/LCD Drive Ripple Voltage				100	[mV] p-p	

Note 1 : PDD(typ.)@ mosaic pattern

Note 2 : With Over driver@ H-stripe ; V_{DD} = Typical



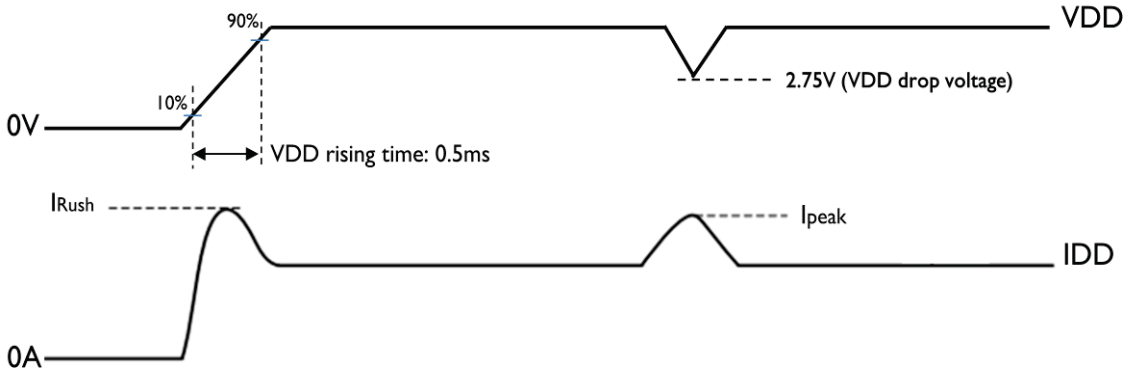
Note 3 :

a. IRush: VDD measured rising timing @ 0.5ms

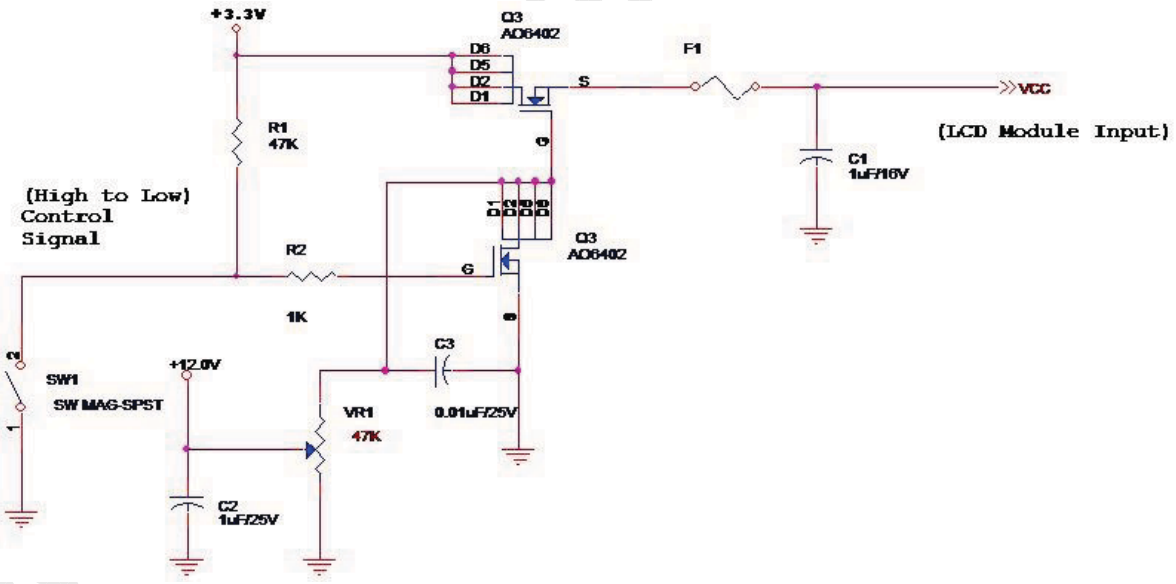
b. IPeak: VDD drop Voltage ≧ 2.75V at this Ipeak current

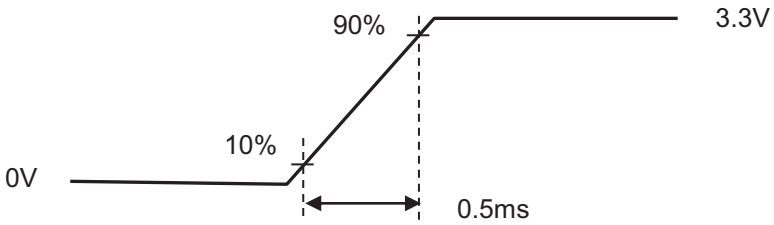
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c. IRush Measure Condition:





Vin rising time

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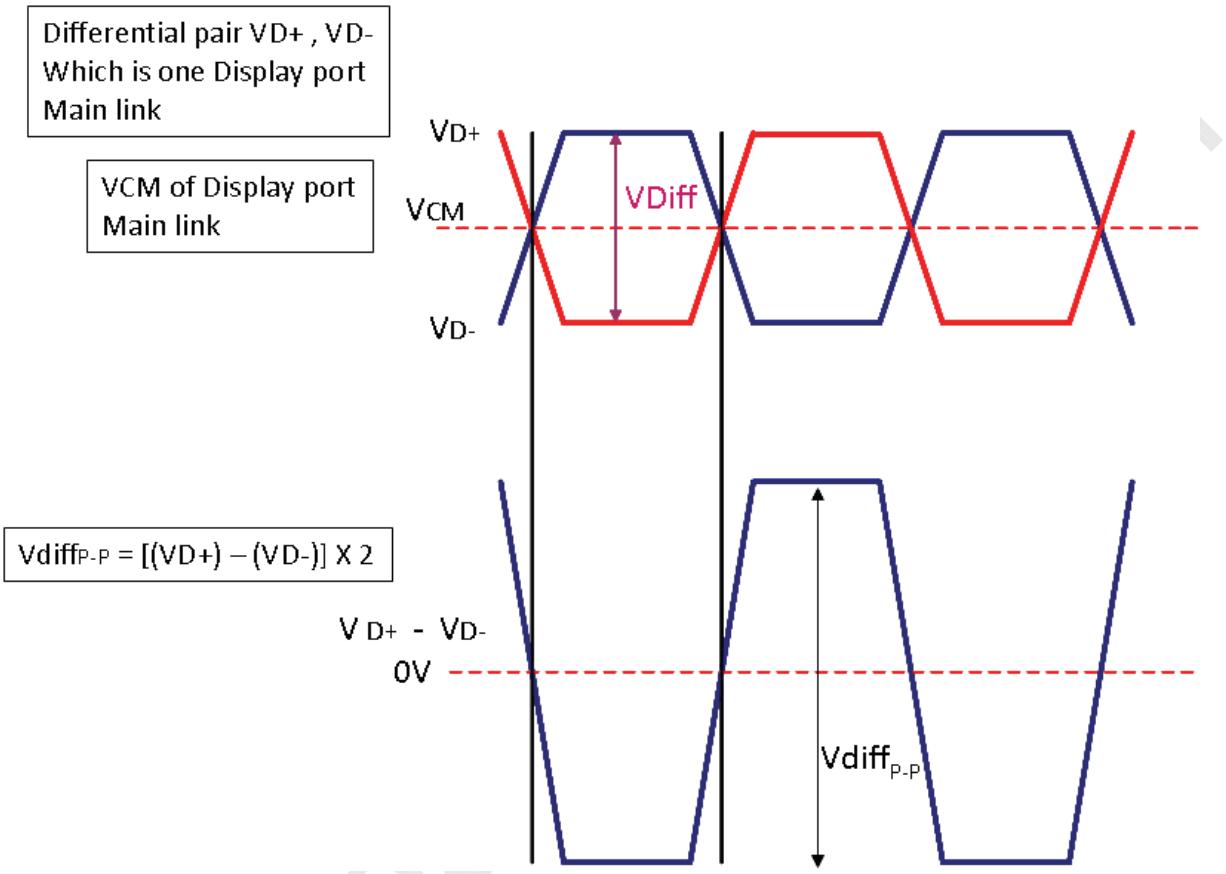
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5.1.2 Signal Electrical Characteristics

Input signals shall be low or High-impedance state when VDD is off.

Signal electrical characteristics are as follows;

Display Port main link signal:



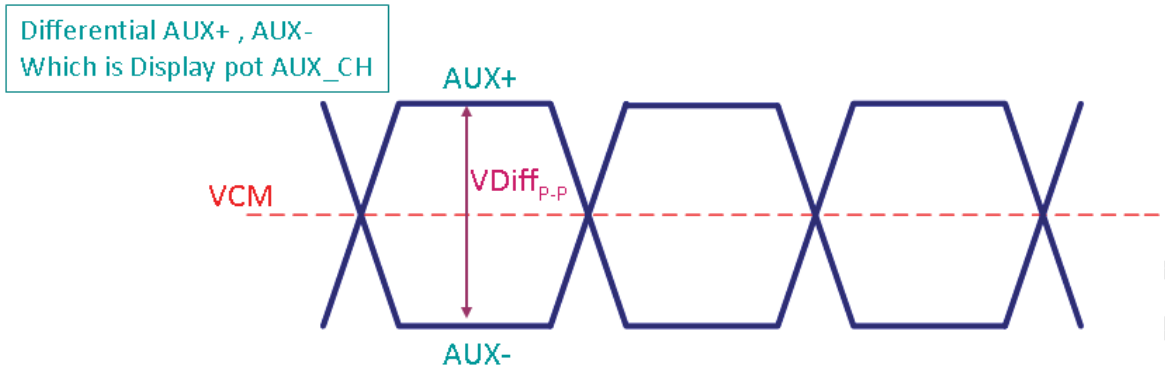
Display port main link					
		Min	Typ	Max	unit
VCM	RX input DC Common Mode Voltage		0		V
$VDiff_{P-P}$	Peak-to-peak Voltage at a receiving Device	HBR3:75		1320	mV

Follow as VESA display port standard V1.4

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Display Port AUX_CH signal:



Display port AUX_CH					
		Min	Typ	Max	unit
VCM	AUX DC Common Mode Voltage		0		V
VDiff P-P	AUX Peak-to-peak Voltage at a receiving Device	270		800	mV

Follow as VESA display port standard V1.3

Display Port VHPD signal:

Display port VHPD					
		Min	Typ	Max	unit
VHPD	HPD Voltage	2.25		3.6	V

Follow as VESA display port standard V1.3

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5.2 Backlight Unit

5.2.1 LED characteristics

Parameter	Symbol	Min	Typ	Max	Units	Condition
Backlight Power Consumption	PLED	-	-	5.6	[Watt]	(Ta=25℃), Note 1 Vin =12V
LED Life-Time	N/A	15,000	-	-	Hour	(Ta=25℃), Note 2 I_F=20 mA

Note 1: Calculator value for reference P_{LED} = VF (Normal Distribution) * IF (Normal Distribution) / Efficiency

Note 2: The LED life-time define as the estimated time to 50% degradation of initial luminous.

5.2.2 Backlight input signal characteristics

Parameter	Symbol	Min	Typ	Max	Units	Remark
LED Power Supply	VLED	6.0	12.0	21.0	[Volt]	Define as Connector Interface (Ta=25℃)
LED Enable Input High Level	VLED_EN	2.5	-	5.5	[Volt]	
LED Enable Input Low Level		-	-	0.5	[Volt]	
PWM Logic Input High Level	VPWM_EN	2.5	-	5.5	[Volt]	
PWM Logic Input Low Level		-	-	0.5	[Volt]	
PWM Input Frequency	FPWM	200	1K	10K	Hz	
PWM Duty Ratio	Duty	5	--	100	%	

Note 1 : Recommend system pull up/down resistor no bigger than 10kohm

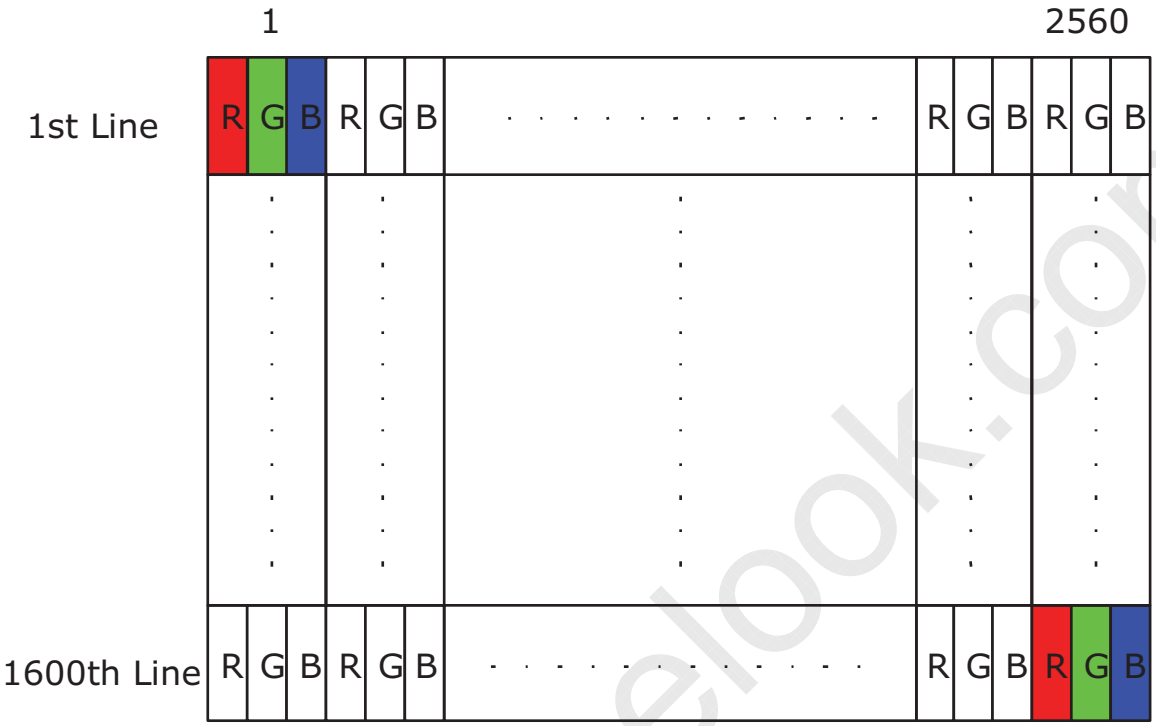
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6. Signal Interface Characteristic

6.1 Pixel Format Image

Following figure shows the relationship of the input signals and LCD pixel format.



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6.2 Integration Interface Requirement

6.2.1 Connector Description

Physical interface is described as for the connector on module.

These connectors are capable of accommodating the following signals and will be following components.

Connector Name / Designation	For Signal Connector
Manufacturer	Starncon
Type / Part Number	Starconn 300E40-1010RA-G3
Mating Housing/Part Number	IPEX 20704-040T-13 or compatible

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6.2.2 Pin Assignment (4 Lane)

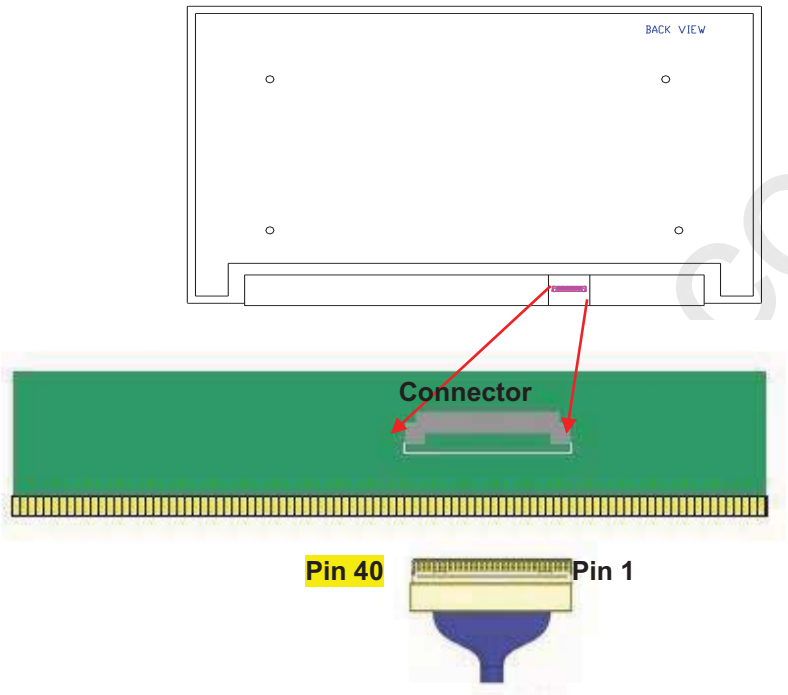
eDP lane is a differential signal technology for LCD interface and high speed data transfer device.

PIN NO	Symbol	Function
1	NC	NC
2	H_GND	High Speed Ground
3	Lane3_N	Comp Signal Lane3
4	Lane3_P	True Signal Link Lane 3
5	H_GND	High Speed Ground
6	Lane2_N	Comp Signal Link Lane 2
7	Lane2_P	True Signal Link Lane 2
8	H_GND	High Speed Ground
9	Lane1_N	Comp Signal Lane 1
10	Lane1_P	True Signal Link Lane 1
11	H_GND	High Speed Ground
12	Lane0_N	Comp Signal Link Lane 0
13	Lane0_P	True Signal Link Lane 0
14	H_GND	High Speed Ground
15	AUX_CH_P	True Signal Auxiliary Ch.
16	AUX_CH_N	Comp Signal Auxiliary Ch.
17	H_GND	High Speed Ground
18	LCD_VCC	LCD logic and driver power
19	LCD_VCC	LCD logic and driver power
20	LCD_VCC	LCD logic and driver power
21	LCD_VCC	LCD logic and driver power
22	LCD Self Test	LCD Panel Self Test Enable
23	LCD_GND	LCD logic and driver ground
24	LCD_GND	LCD logic and driver ground
25	LCD_GND	LCD logic and driver ground
26	LCD_GND	LCD logic and driver ground
27	HPD	HPD signal pin
28	BL_GND	Backlight_ground
29	BL_GND	Backlight_ground
30	BL_GND	Backlight_ground
31	BL_GND	Backlight_ground
32	BL_Enable	Backlight On / Off
33	BL_PWM_DIM	System PWM signal Input
34	NC	NC
35	NC	NC
36	BL_PWR	Backlight power

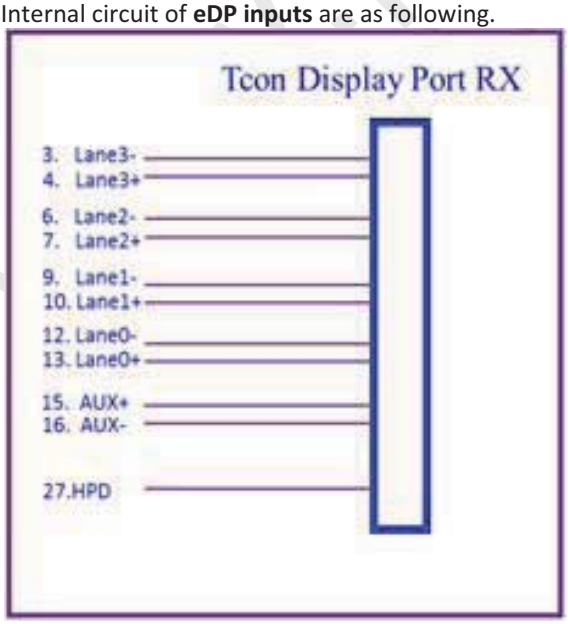
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37	BL_PWR	Backlight power
38	BL_PWR	Backlight power
39	BL_PWR	Backlight power
40	OD	OD (Enable @ Low, 0V~0.36V); (Disable @ High, 1.71V~1.89V)(Note3)



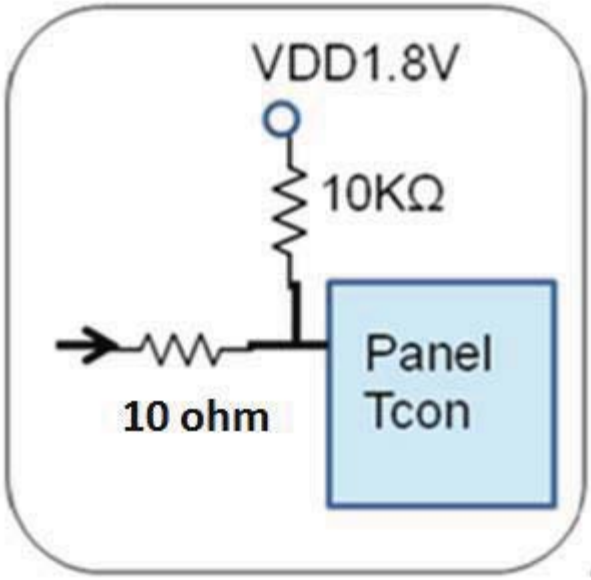
Note1: Start from right side.
Note2: Input signals shall be low or High-impedance state when VDD is off.
Internal circuit of **eDP inputs** are as following.



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Note3: Input equivalent impedance of color matrix/ over driver pin.



6.3 Interface Timing

6.3.1 Timing Characteristics

For normal display, interface timings should match the 2560x1600 /165Hz manufacturing guide line timing.

Parameter		Symbol	Min.	Typ.	Max.	Unit
Frame Rate		-	60		165	Hz
Clock frequency		1/ T_{Clock}	282.8		777.5	MHz
Vertical Section	Period	T_V	1732		1732	T_{Line}
	Active	T_{VD}	1600			
	Blanking	T_{VB}	132		132	
Horizontal Section	Period	T_H	2720		2720	T_{Clock}
	Active	T_{HD}	2560			
	Blanking	T_{HB}	160		160	

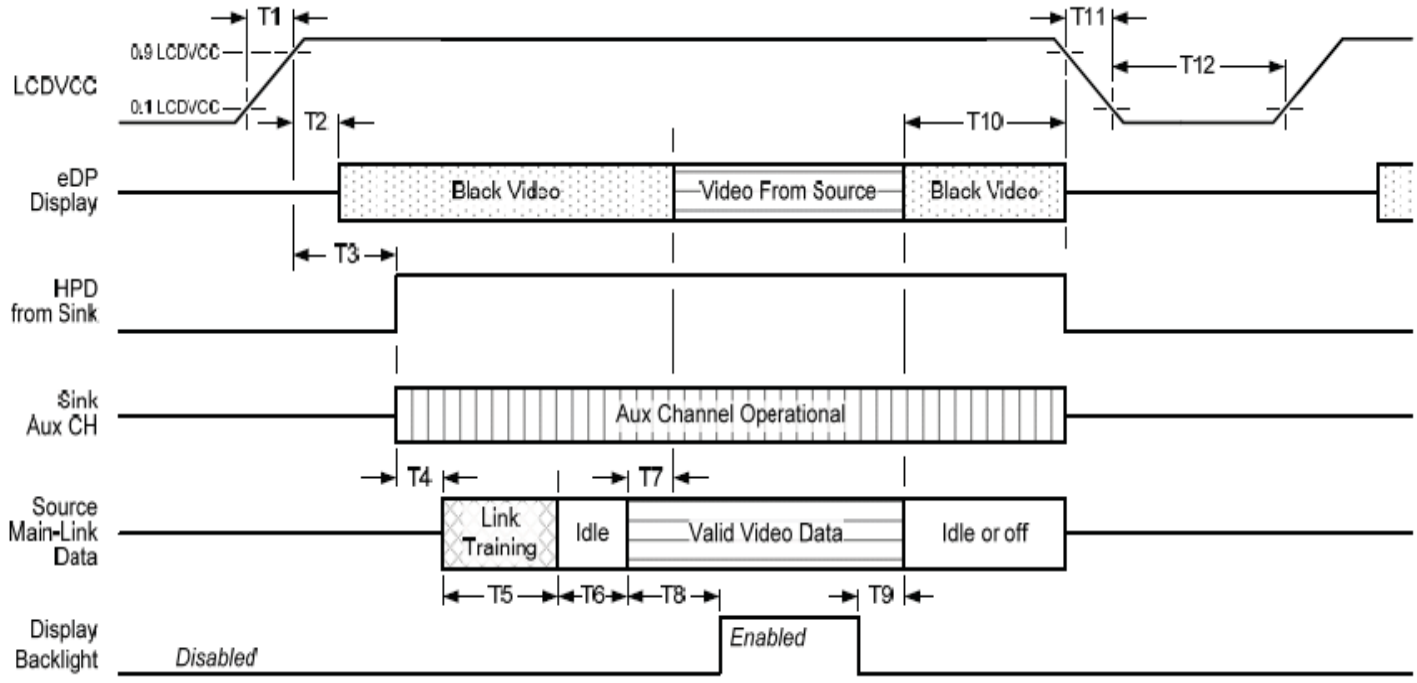
Note 1 : G Sync function enble frame rate frequency operating range: 48-165Hz

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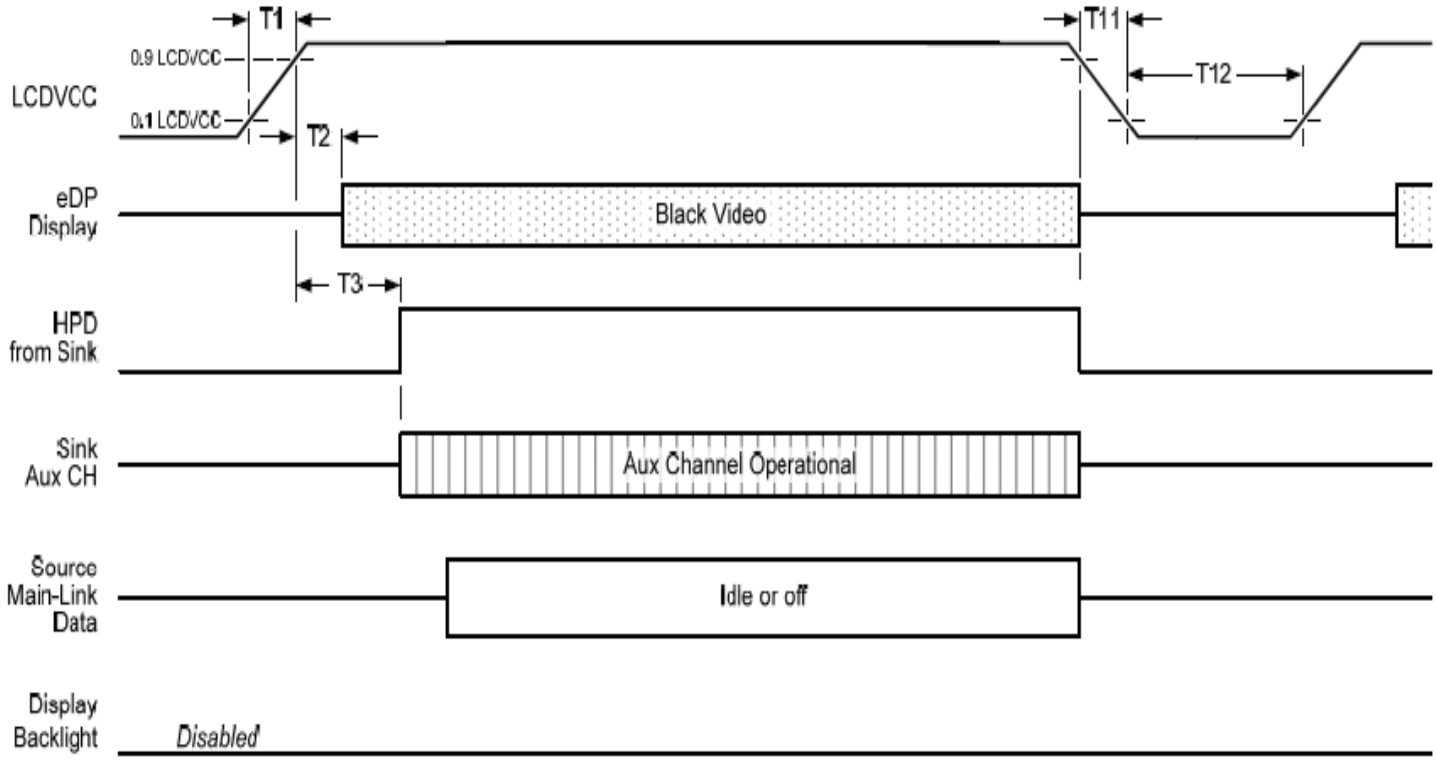
6.4 Power ON/OFF Sequence

Display Port panel power sequence:



Display port interface power up/down sequence, normal system operation

Display Port AUX_CH transaction only:



Display port interface power up/down sequence, AUX_CH transaction only

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Display Port panel power sequence timing parameter:

Timing parameter	Description	Reqd. by	Limits			Notes
			Min.	Typ.	Max.	
T1	power rail rise time, 10% to 90%	source	0.5ms		10ms	
T2	delay from LCDVDD to black video generation	sink	0ms		200ms	prevents display noise until valid video data is received from the source
T3	delay from LCDVDD to HPD high	sink	0ms		200ms	sink AUX_CH must be operational upon HPD high.
T4	delay from HPD high to link training initialization	source				allows for source to read link capability and initialize.
T5	link training duration	source				dependant on source link to read training protocol.
T6	link idle	source				Min accounts for required BS-Idle pattern. Max allows for source frame synchronization.
T7	delay from valid video data from source to video on display	sink	0ms		50ms	max allows sink validate video data and timing.
T8	delay from valid video data from source to backlight enable	source				source must assure display video is stable.
T9	delay from backlight disable to end of valid video data	source				source must assure backlight is no longer illuminated.
T10	delay from end of valid video data from source to power off	source	50ms		500ms	
T11	power rail fall time, 905 to 10%	source			10ms	
T12	power off time	source	500ms			

Note1: The sink must include the ability to generate black video autonomously. The sink must automatically enable black video under the following conditions:
-upon LCDVDD power on (with in T2 max)-when the "Novideostream_Flag" (VB-ID Bit 3) is received from the source (at the end of T9).
-When no main link data or invalid video data is received from the source. Black video must be displayed within 64ms (typ) from the start of either condition. Video data can be deemed invalid based on MSA and timing information, for example.

Note 2: The sink may implement the ability to disable the black video function, as described in Note 1, above, for system development and debugging purpose.

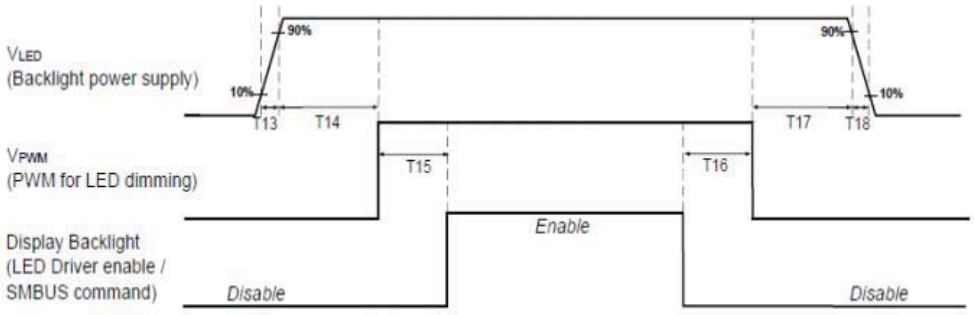
Note 3: The sink must support AUX_CH polling by the source immediately following LCDVDD power on without causing damage to the sink device (the source can re-try if the sink is not ready). The sink must be able to respond to an AUX_CH transaction with the time specified within T3 max.

Note 4:T8>T7

Display Port panel B/L power sequence timing parameter:

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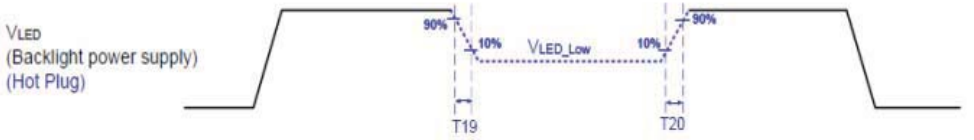


	Min (ms)	Max (ms)
T13	0.5	10
T14	10	-
T15	0	-
T16	0	-
T17	10	-
T18	0.5	10
T19	1*	-
T20	1*	-

Seamless change: T19/T20 = 5xT_{PWM}*

*T_{PWM}= 1/PWM Frequency

Note : When the adapter is hot plugged, the backlight power supply sequence is shown as below.



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7. Panel Reliability Test

7.1 Vibration Test

Test Spec:

- Test method: Non-Operation
- Acceleration: 1.5 G
- Frequency: 10 - 500Hz Random
- Sweep: 30 Minutes each Axis (X, Y, Z)

7.2 Shock Test

Test Spec:

- Test method: Non-Operation
- Acceleration: 220 G , Half sine wave
- Active time: 2 ms
- Pulse: X,Y,Z .one time for each side

7.3 Reliability Test

Items	Required Condition	Note
Temperature Humidity Bias	Ta= 40℃, 90%RH, 300h	
High Temperature Operation	Ta= 50℃, Dry, 300h	
Low Temperature Operation	Ta= 0℃, 300h	
High Temperature Storage	Ta= 60℃, 35%RH, 300h	
Low Temperature Storage	Ta= -20℃, 50%RH, 250h	
Thermal Shock Test	Ta=-20℃to 60℃, Duration at 30 min, 100 cycles	
ESD	Contact : ±8 KV Air : ±15 KV	Note 1

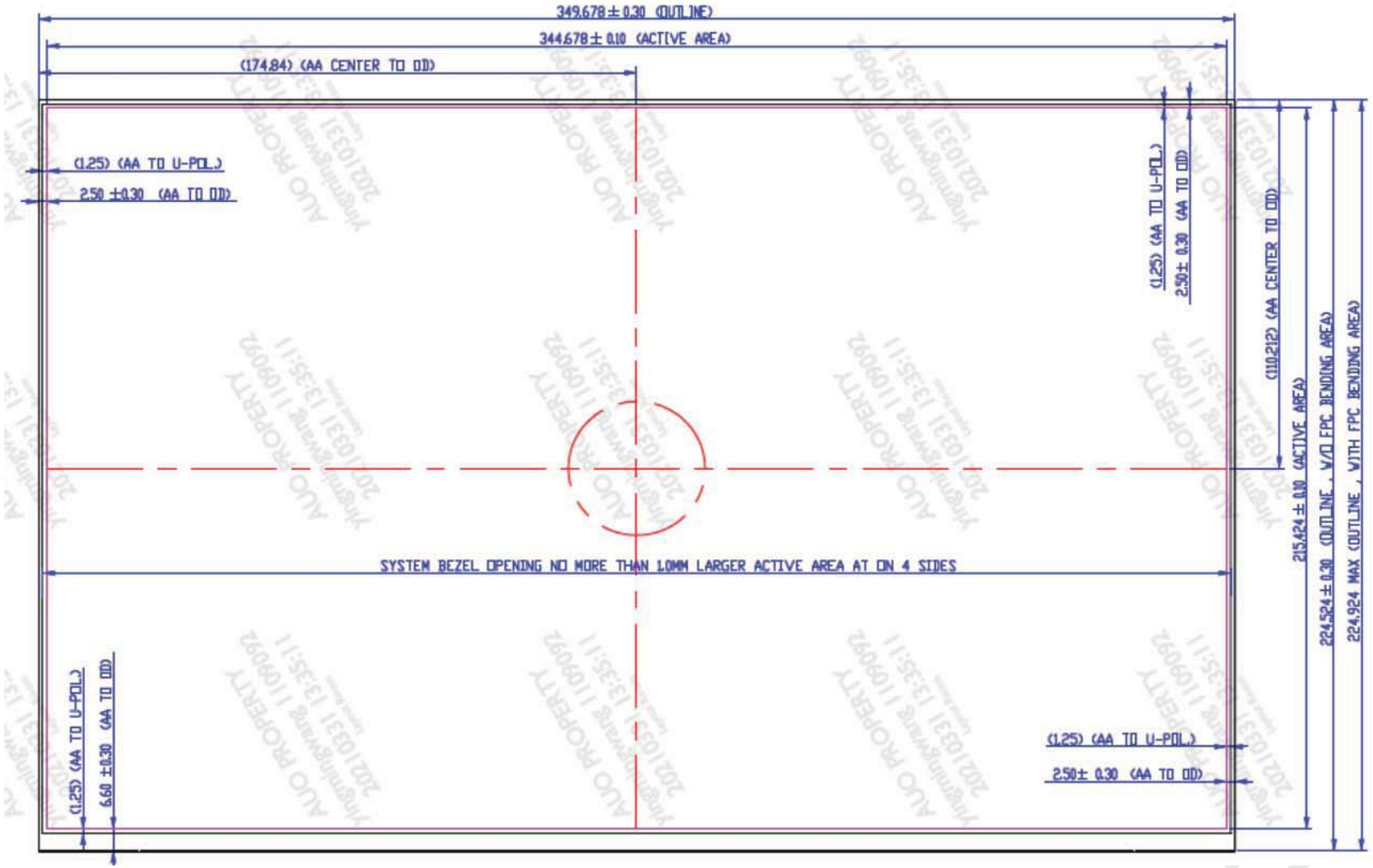
Note1: According to EN 61000-4-2 , ESD class B: Some performance degradation allowed. Self-recoverable, No data lost, No hardware failures.

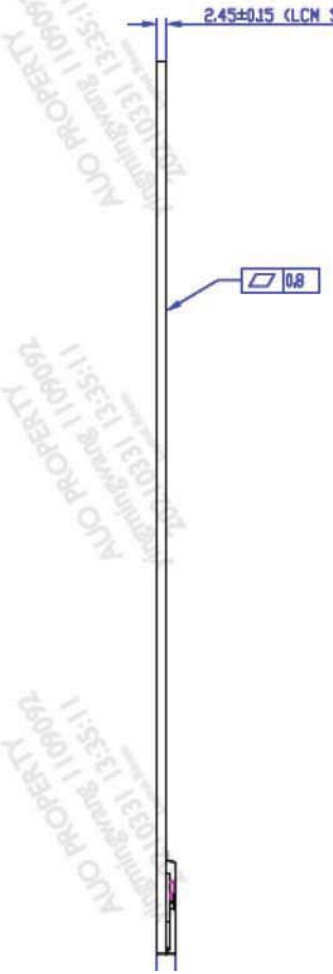
Remark: MTBF (Excluding the LED): 30,000 hours with a confidence level 90%

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8. Mechanical Characteristics

8.1 LCM Outline Dimension



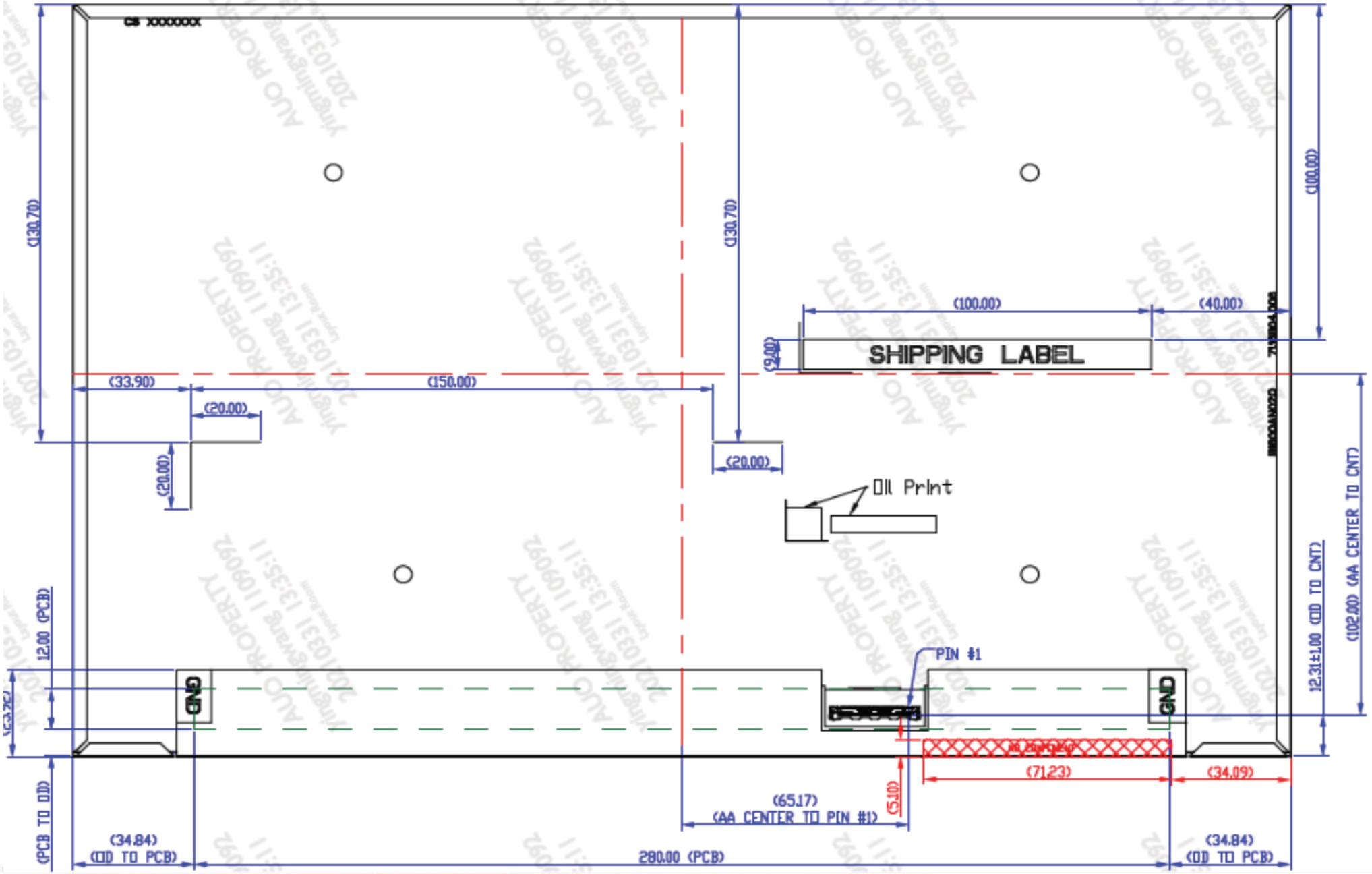


Note: Prevention IC damage, IC positions not allowed any overlap over these areas.

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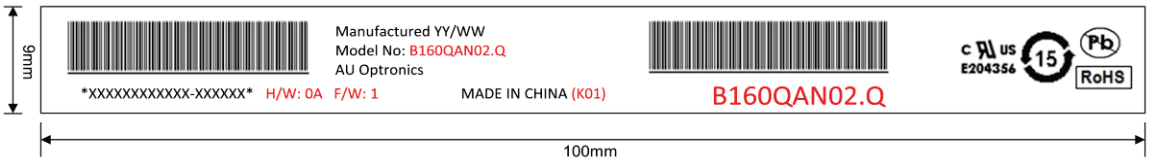
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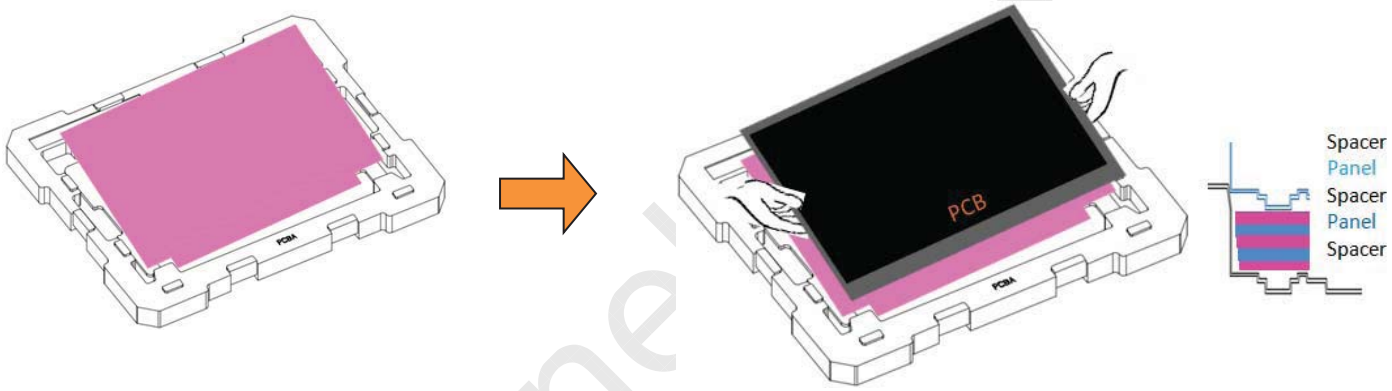
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9. Shipping and Package

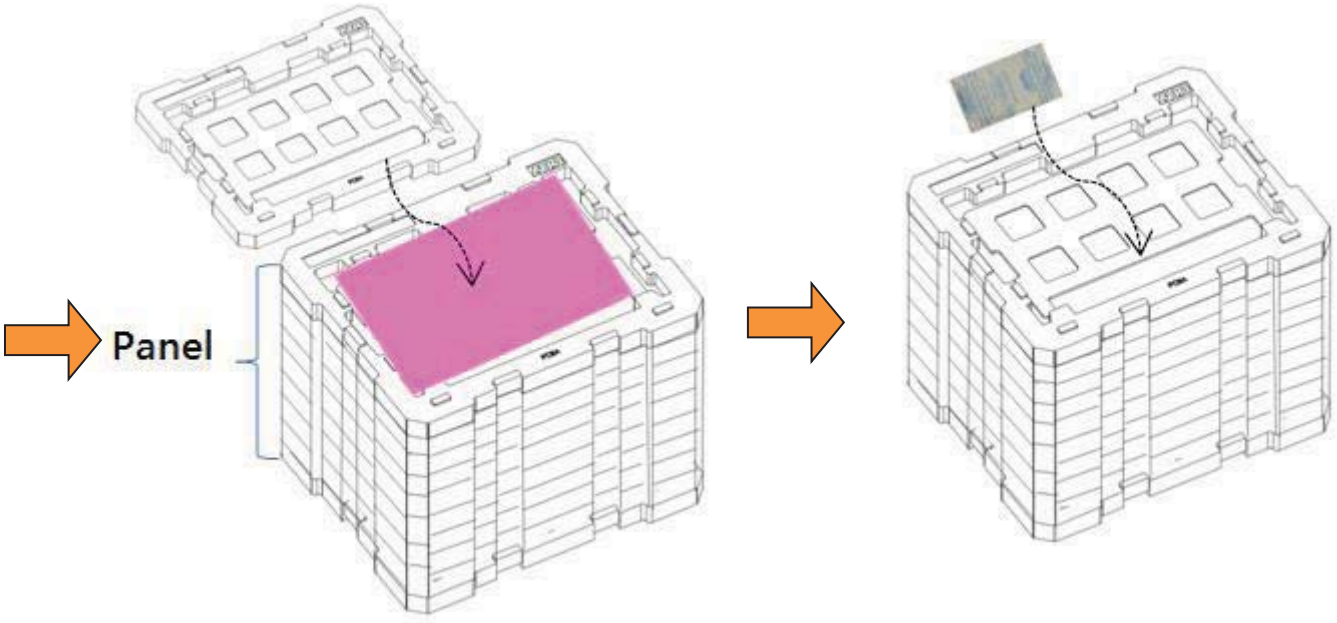
9.1 Shipping Label Format



9.2 Carton Package



1 TRAY with 2 pcs Panels and 3 pcs Spacer

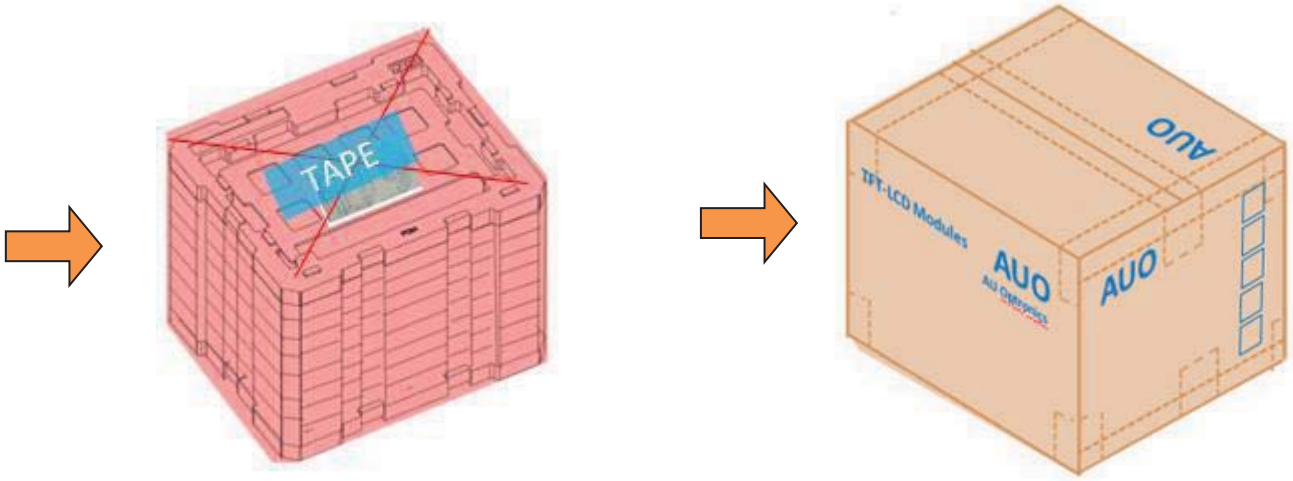


Packing 12 layers TRAY with Panel and spacer,
And then top cover 1 empty TRAY
Total pack 24 pcs Panels.

Put 1bag distance bag

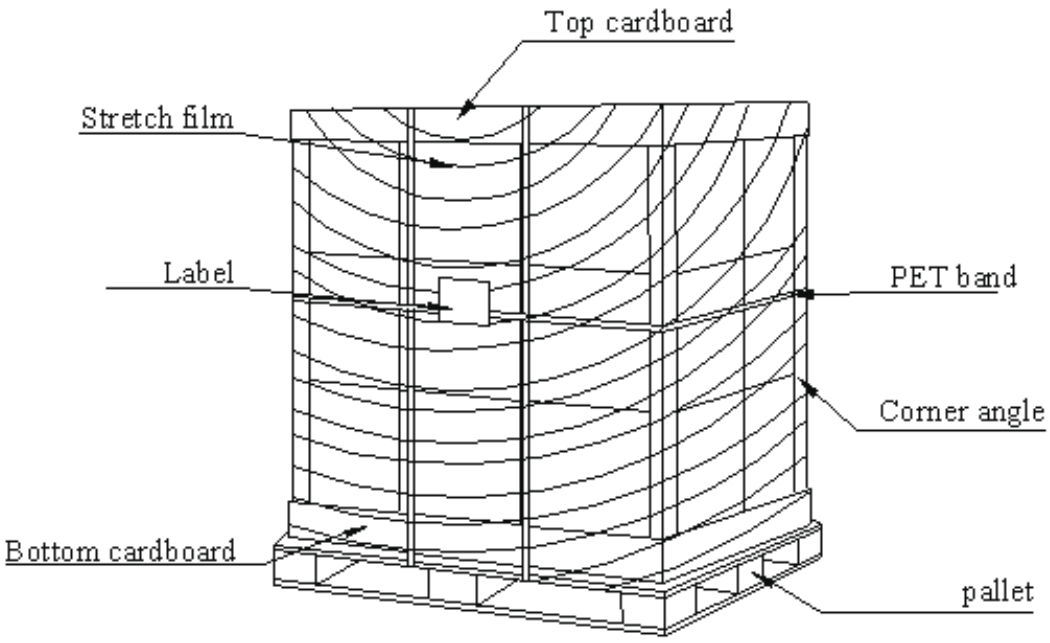
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24 pcs Panel/carton

9.3 Shipping Package of Palletizing Sequence



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10. Appendix:

10.1 EDID Description

B160QAN02 Q EDID Code

Address HEX	FUNCTION	Value HEX	Value BIN	Value DEC	Note
00	Header	00	00000000	0	
01	Header	FF	11111111	255	
02	Header	FF	11111111	255	
03	Header	FF	11111111	255	
04	Header	FF	11111111	255	
05	Header	FF	11111111	255	
06	Header	FF	11111111	255	
07	Header	00	00000000	0	
08	EISA Manuf. Code LSB	06	00000110	6	
09	Compressed ASCII	AF	10101111	175	
0A	Product Code	99	10011001	153	
0B	Product Code	C1	11000001	193	
0C	32-bit ser #	00	00000000	0	
0D	ID S/N - option	00	00000000	0	
0E	ID S/N - option	00	00000000	0	
0F	ID S/N - option	00	00000000	0	
10	Week of manufacture	07	00000111	7	
11	Year of manufacture	1F	00011111	31	
12	EDID Structure Ver.	01	00000001	1	
13	EDID revision #	04	00000100	4	
14	**Video input def. (digital I/P, non-TMDS, CRGB)**	A5	10100101	165	
15	**Max H image size (rounded to cm)**	22	00100010	34	
16	**Max V image size (rounded to cm)**	16	00010110	22	
17	**Display Gamma (=(gamma*100)-100)**	78	01111000	120	
18	**Feature support (no DPMS, Active OFF, RGB, tmg Blk#1)**	03	00000011	3	
19	Red/green low bits (Lower 2:2:2:2 bits)	C2	11000010	194	
1A	Blue/white low bits (Lower 2:2:2:2 bits)	D5	11010101	213	
1B	Red x (Upper 8 bits)	AD	10101101	173	
1C	Red y/ highER 8 bits	51	01010001	81	
1D	Green x	42	01000010	66	
1E	Green y	B1	10110001	177	
1F	Blue x	25	00100101	37	
20	Blue y	0D	00001101	13	

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21	White x	50	01010000	80	
22	White y	54	01010100	84	
23	Established timing 1	00	00000000	0	
24	Established timing 2	00	00000000	0	
25	Established timing 3	00	00000000	0	
26	Standard timing #1	01	00000001	1	
27	Standard timing #1	01	00000001	1	
28	Standard timing #2	01	00000001	1	
29	Standard timing #2	01	00000001	1	
2A	Standard timing #3	01	00000001	1	
2B	Standard timing #3	01	00000001	1	
2C	Standard timing #4	01	00000001	1	
2D	Standard timing #4	01	00000001	1	
2E	Standard timing #5	01	00000001	1	
2F	Standard timing #5	01	00000001	1	
30	Standard timing #6	01	00000001	1	
31	Standard timing #6	01	00000001	1	
32	Standard timing #7	01	00000001	1	
33	Standard timing #7	01	00000001	1	
34	Standard timing #8	01	00000001	1	
35	Standard timing #8	01	00000001	1	
36	Pixel Clock/10000 LSB	78	01111000	120	
37	Pixel Clock/10000 USB	6E	01101110	110	
38	Horz active Lower 8bits	00	00000000	0	
39	Horz blanking Lower 8bits	A0	10100000	160	
3A	HorzAct:HorzBlnk Upper 4:4 bits	A0	10100000	160	
3B	Vertical Active Lower 8bits	40	01000000	64	
3C	Vertical Blanking Lower 8bits	84	10000100	132	
3D	Vert Act : Vertical Blanking (upper 4:4 bit)	60	01100000	96	
3E	HorzSync. Offset	30	00110000	48	
3F	HorzSync.Width	20	00100000	32	
40	VertSync.Offset : VertSync.Width	AA	10101010	170	
41	Horz&Vert Sync Offset/Width Upper 2bits	00	00000000	0	
42	Horizontal Image Size Lower 8bits	58	01011000	88	
43	Vertical Image Size Lower 8bits	D7	11010111	215	
44	Horizontal & Vertical Image Size (upper 4:4 bits)	10	00010000	16	
45	Horizontal Border (zero for internal LCD)	00	00000000	0	
46	Vertical Border (zero for internal LCD)	00	00000000	0	
47	Signal (non-intr, norm, no stero, sep sync, neg pol)	18	00011000	24	

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48	Pixel Clock/10000 LSB	00	00000000	0	
49	Pixel Clock/10000 USB	00	00000000	0	
4A	Horz active Lower 8bits	00	00000000	0	
4B	Horz blanking Lower 8bits	FD	11111101	253	
4C	HorzAct:HorzBlnk Upper 4:4 bits	0C	00001100	12	
4D	Vertical Active Lower 8bits	30	00110000	48	
4E	Vertical Blanking Lower 8bits	A5	10100101	165	
4F	Vert Act : Vertical Blanking (upper 4:4 bit)	1F	00011111	31	
50	HorzSync. Offset	1F	00011111	31	
51	HorzSync.Width	4E	01001110	78	
52	VertSync.Offset : VertSync.Width	01	00000001	1	
53	Horz&Vert Sync Offset/Width Upper 2bits	0A	00001010	10	
54	Horizontal Image Size Lower 8bits	20	00100000	32	
55	Vertical Image Size Lower 8bits	20	00100000	32	
56	Horizontal & Vertical Image Size (upper 4:4 bits)	20	00100000	32	
57	Horizontal Border (zero for internal LCD)	20	00100000	32	
58	Vertical Border (zero for internal LCD)	20	00100000	32	
59	Signal (non-intr, norm, no stero, sep sync, neg pol)	20	00100000	32	
5A	descriptor #3	00	00000000	0	
5B	Reserved for definition	00	00000000	0	
5C	Reserved for definition	00	00000000	0	
5D	ASCII String	FE	11111110	254	
5E	Reserved for definition	00	00000000	0	
5F	Manufacture	41	01000001	65	A
60	Manufacture	55	01010101	85	U
61	Manufacture	4F	01001111	79	O
62	Reserved for definition	0A	00001010	10	
63	Reserved for definition	20	00100000	32	
64	Reserved for definition	20	00100000	32	
65	Reserved for definition	20	00100000	32	
66	Reserved for definition	20	00100000	32	
67	Reserved for definition	20	00100000	32	
68	Reserved for definition	20	00100000	32	
69	Reserved for definition	20	00100000	32	
6A	Reserved for definition	20	00100000	32	
6B	Reserved for definition	20	00100000	32	
6C	Reserved for definition	00	00000000	0	
6D	descriptor #4	00	00000000	0	
6E	Reserved for definition	00	00000000	0	

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6F	Reserved for definition	FE	11111110	254	
70	Reserved for definition	00	00000000	0	
71	Manufacture P/N	42	01000010	66	B
72	Manufacture P/N	31	00110001	49	1
73	Manufacture P/N	36	00110110	54	6
74	Manufacture P/N	30	00110000	48	0
75	Manufacture P/N	51	01010001	81	Q
76	Manufacture P/N	41	01000001	65	A
77	Manufacture P/N	4E	01001110	78	N
78	Manufacture P/N	30	00110000	48	0
79	Manufacture P/N	32	00110010	50	2
7A	Manufacture P/N	2E	00101110	46	.
7B	Manufacture P/N	51	01010001	81	Q
7C	Reserved for definition	20	00100000	32	
7D	Reserved for definition	0A	00001010	10	
7E	Extension Flag	01	00000001	1	
7F	Checksum	66	01100110	102	

EXTENSION EDID Code for HF

Address	FUNCTION	Value	Value	Value	Note
HEX		HEX	BIN	DEC	
80		70	01110000	112	
81	Display ID Version	13	00010011	19	
82	Block Size	79	01111001	121	
83		00	00000000	0	
84		00	00000000	0	
85	Type I timing	03	00000011	3	
86	Revision '1'	01	00000001	1	
87	Number of Payload Bytes in Block	14	00010100	20	
88	Pixel clock	B6	10110110	182	
89	Pixel clock	2F	00101111	47	
8A	Pixel clock	01	00000001	1	
8B	Preferred timing	85	10000101	133	
8C	H-AA	FF	11111111	255	
8D	H-AA	09	00001001	9	
8E	H-BK	9F	10011111	159	
8F	H-BK	00	00000000	0	
90	H-offset	2F	00101111	47	
91	H-offset	00	00000000	0	
92	H-Sync	1F	00011111	31	
93	H-Sync	00	00000000	0	
94	V-AA	3F	00111111	63	
95	V-AA	06	00000110	6	

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96	V-BK	83	10000011	131	
97	V-BK	00	00000000	0	
98	V-offset	09	00001001	9	
99	V-offset	00	00000000	0	
9A	V-Sync	09	00001001	9	
9B	V-Sync	00	00000000	0	
9C		00	00000000	0	
9D		00	00000000	0	
9E		00	00000000	0	
9F		00	00000000	0	
A0		00	00000000	0	
A1		00	00000000	0	
A2		00	00000000	0	
A3		00	00000000	0	
A4		00	00000000	0	
A5		00	00000000	0	
A6		00	00000000	0	
A7		00	00000000	0	
A8		00	00000000	0	
A9		00	00000000	0	
AA		00	00000000	0	
AB		00	00000000	0	
AC		00	00000000	0	
AD		00	00000000	0	
AE		00	00000000	0	
AF		00	00000000	0	
B0		00	00000000	0	
B1		00	00000000	0	
B2		00	00000000	0	
B3		00	00000000	0	
B4		00	00000000	0	
B5		00	00000000	0	
B6		00	00000000	0	
B7		00	00000000	0	
B8		00	00000000	0	
B9		00	00000000	0	
BA		00	00000000	0	
BB		00	00000000	0	
BC		00	00000000	0	
BD		00	00000000	0	
BE		00	00000000	0	
BF		00	00000000	0	
C0		00	00000000	0	
C1		00	00000000	0	
C2		00	00000000	0	

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C3		00	00000000	0	
C4		00	00000000	0	
C5		00	00000000	0	
C6		00	00000000	0	
C7		00	00000000	0	
C8		00	00000000	0	
C9		00	00000000	0	
CA		00	00000000	0	
CB		00	00000000	0	
CC		00	00000000	0	
CD		00	00000000	0	
CE		00	00000000	0	
CF		00	00000000	0	
D0		00	00000000	0	
D1		00	00000000	0	
D2		00	00000000	0	
D3		00	00000000	0	
D4		00	00000000	0	
D5		00	00000000	0	
D6		00	00000000	0	
D7		00	00000000	0	
D8		00	00000000	0	
D9		00	00000000	0	
DA		00	00000000	0	
DB		00	00000000	0	
DC		00	00000000	0	
DD		00	00000000	0	
DE		00	00000000	0	
DF		00	00000000	0	
E0		00	00000000	0	
E1		00	00000000	0	
E2		00	00000000	0	
E3		00	00000000	0	
E4		00	00000000	0	
E5		00	00000000	0	
E6		00	00000000	0	
E7		00	00000000	0	
E8		00	00000000	0	
E9		00	00000000	0	
EA		00	00000000	0	
EB		00	00000000	0	
EC		00	00000000	0	
ED		00	00000000	0	
EE		00	00000000	0	
EF		00	00000000	0	

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F0		00	00000000	0	
F1		00	00000000	0	
F2		00	00000000	0	
F3		00	00000000	0	
F4		00	00000000	0	
F5		00	00000000	0	
F6		00	00000000	0	
F7		00	00000000	0	
F8		00	00000000	0	
F9		00	00000000	0	
FA		00	00000000	0	
FB		00	00000000	0	
FC		00	00000000	0	
FD		00	00000000	0	
FE	Section Checksum (81~FE)	22	100010	34	
FF	EDID Extension Block Checksum (80~FF)	90	10010000	144	

10.2 Notes

Item	DPCD Ver.	eDP	LRR	DRRS (Static DRRS)	sDRRS (Seamless DRRS)	DCR	DMRRS	PSR
Panel	1.4	1.4	NA	Off	Off	Off	Off	PSR1

MBO	VESA DSC	MSO	AMD	Intel	NVidia	HDR
			Free-Sync	Adaptive Sync	G-Sync	
Off	Off	Off	On	On	On	NA

Dimming	Timing 1	Timing 2	Timing 3
PWM	165	60	48-165Hz

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